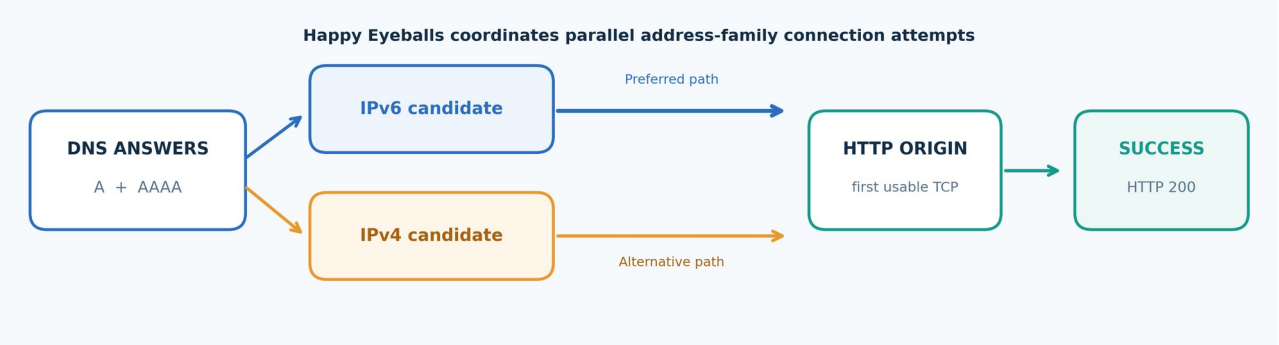
AIORI 3 | INTERNET MEASUREMENT

PS-028

Happy Eyeballs IPv4/IPv6 Fallback Measurement

A controlled, packet-validated comparison of dual-stack browser connection behaviour



Happy Eyeballs coordinates IPv6 and IPv4 candidate paths toward the first successful connection.

TEAM ALGORHYTHM

Campaign: ps028-a10-live-20261030-01
280 valid live trials | 7 conditions | 2 browsers | 20 repetitions per cell
Report snapshot: 31 October 2026

Evidence baseline

Repository branch a/lab-foundation; code and evidence commit 002c4453d35d6760760c6a14d2e98618b024f1f2. Prepared from the campaign manifest, per-trial records, 280 PCAPs, browser telemetry, checksum manifests, analysis outputs and recorded test log.

1. Abstract (Problem Description)

In dual-stack networks, a hostname can resolve to both IPv6 (AAAA) and IPv4 (A) addresses while the corresponding paths differ in reachability or latency. If a client waits too long for an impaired preferred path, connection establishment can be delayed. Happy Eyeballs addresses this by coordinating connection attempts across available endpoints so a working path can be selected without waiting for a long timeout.

This study evaluates Google Chrome 156.0.8078.73 and Mozilla Firefox 159.0 in an isolated Linux dual-stack laboratory. Six impairment scenarios required by PS-028 were tested alongside a healthy baseline control. Each condition was repeated 20 times for each browser, yielding 280 valid live trials. The campaign ledger records 282 attempts because two initial cleanup failures were preserved and retried.

Principal result

Across the 280 primary document navigations, 154 (55.00%) completed over IPv6 and 126 (45.00%) completed over IPv4 after an IPv6 attempt. All 280 valid trials returned HTTP 200. Raw-PCAP flow correlation found the primary GET / TCP flow in all 280 captures and no disagreement with the browser-recorded winning address family.

The report separates the primary document navigation from later ancillary or speculative connections. This distinction matters in S5: an IPv4 favicon request or speculative preconnect may appear after a successful IPv6 main navigation and must not be mistaken for a fallback of the document request.

Scope is deliberately narrow: two recorded browser builds, one host/testbed environment, HTTP/1.1 over TCP, and the scenarios in the canonical `lab/scenarios.yaml`. Findings are empirical and build-specific; they are not general RFC conformance verdicts.

2. Background

2.1 Dual-stack connection establishment

A DNS lookup can return multiple destination addresses. RFC 6724 describes default address selection and the address-ordering context; Happy Eyeballs adds an application connection-establishment strategy that avoids serially waiting for a broken address family. RFC 8305 describes asynchronous hostname resolution, address sorting and staggered connection attempts. When a connection succeeds, outstanding attempts should be cancelled or abandoned as appropriate.

2.2 What is measured in this experiment

- Winning family: the address family used by the TCP flow carrying the primary HTTP GET / document request that receives HTTP 200.
- Fallback: the primary navigation starts on IPv6 but its successful document request is carried over IPv4. A separate IPv4 ancillary request after an IPv6 document success is not a main-navigation fallback.
- Primary-flow timing: raw-PCAP interval from the client SYN to the matching server SYN/ACK on the same TCP stream as the primary GET / request. It is a packet-derived connection-response/RTT proxy, not full page-load time.
- Inter-family first-SYN interval: time between first observed IPv6 and IPv4 SYNs. It is useful for observing pacing, but it is not automatically fallback delay when the IPv4 flow is ancillary or speculative.

2.3 Research framing

Prior work has shown that clients and resolvers differ in their Happy Eyeballs timing and resolution behaviour. This study contributes a reproducible controlled matrix and raw packet evidence for two pinned browser builds, rather than proposing a new browser stack or inferring internal algorithms from undocumented telemetry alone.

3. Literature and RFC Review

Reference	Role in this study
RFC 8305 (2017), Happy Eyeballs v2	Primary normative baseline for DNS resolution handling (Section 3), address sorting (Section 4), and connection attempts (Section 5).
RFC 6724, Default Address Selection	Context for address ordering and IPv6/IPv4 preference on dual-stack hosts.
RFC 6555, Happy Eyeballs v1	Historical predecessor; obsoleted by RFC 8305.
RFC 6556, Testing Eyeball Happiness	Informational measurement methodology for externally observed session setup and packet-level metrics.
draft-ietf-happy-happyeyeballs-v3-04	Pinned evolving reference, 2 July 2026; active Internet-Draft and not an RFC as of 31 October 2026.
Lazy Eye Inspection (ACM IMC 2025)	Related empirical study of Happy Eyeballs implementations, browsers and resolvers; motivation for observing real client behaviour.
IETF 126 WebTester materials / source	Measurement-tooling context and reference implementation material, not a normative standard.

3.1 Relevant RFC 8305 parameters

- Resolution Delay (Section 3): when A arrives before AAAA, RFC 8305 recommends a 50 ms wait for the preferred AAAA result. If a positive AAAA answer has not arrived by the end of that interval, the client SHOULD proceed using available IPv4 addresses while continuing the connection procedure.
- Connection attempts (Section 5): a subsequent attempt MUST NOT start within 10 ms of the previous attempt. The recommended minimum Connection Attempt Delay is 100 ms; the recommended default is 250 ms; the recommended maximum is 2 seconds.
- Active failure: Section 5 describes advancing to another candidate when a connection attempt fails; measurements here observe the clients' response to a TCP RST rather than assuming their internal failure state machine.

Standards interpretation

The S2 Chrome measurements are important because every recorded first inter-family interval in that cell is below 10 ms. This is reported as a packet-observed result for the tested Chrome build and environment. The project does not infer that all Chrome versions are non-conforming.

4. Proposed Solution

4.1 Proposed workflow and controlled testbed

The lab isolates the client, forwarding/DNS controller and origin server in three Linux network namespaces. The client namespace carries dual-stack traffic and runs the browser under an ephemeral profile. The router namespace forwards IPv4/IPv6, hosts the controlled DNS service, and applies scenario-specific nftables rules and traffic-control delays. The server namespace runs Nginx as an HTTP/1.1 origin on both families.

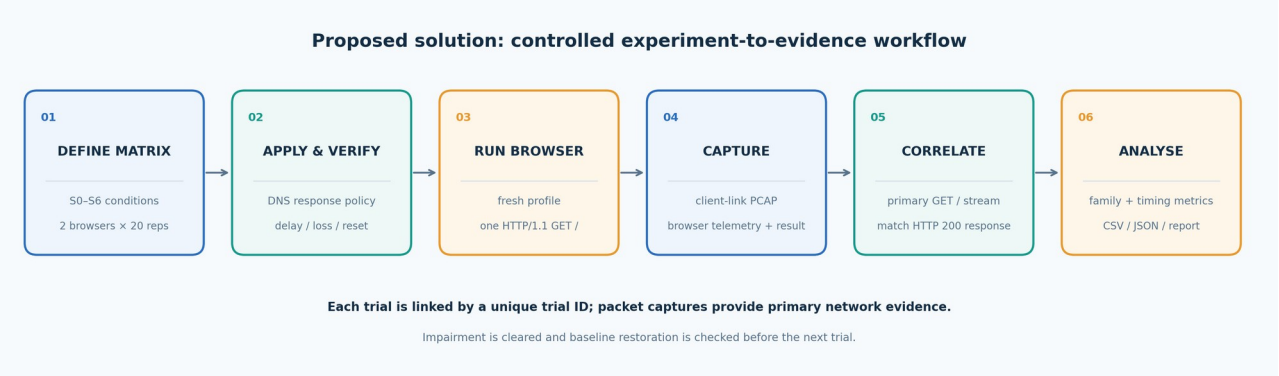


Figure 1. Proposed measurement workflow from controlled scenario definition to reproducible analysis.

4.2 Scenario matrix

ID	Canonical scenario	Applied condition	Expected stress
S0	healthy_baseline	20 ms IPv4 and IPv6 RTT; DNS A/AAAA immediate	Reference behaviour; healthy IPv6 should be preferred.
S1	ipv6_syn_blackhole	Silently drop IPv6 TCP SYN	Measure IPv4 recovery after an unresponsive IPv6 path.
S2	ipv6_tcp_rst	Return TCP RST for IPv6 SYN	Measure response to active IPv6 connection rejection.
S3	ipv6_rtt_150ms	IPv6 RTT approximately 150 ms; IPv4 approximately 20 ms	Measure whether IPv6 completes before alternate-family attempt.
S4	ipv6_rtt_600ms	IPv6 RTT approximately 600 ms; IPv4 approximately 20 ms	Test whether the faster IPv4 path wins.
S5	ipv6_syn_loss_50pct	Approximately 50% loss of client-to-server IPv6 SYN only	Observe stochastic IPv6 completion and alternate-family racing.
S6	delay_aaaa_500ms	A immediate; AAAA delayed 500 ms	Observe behaviour when A precedes a delayed AAAA response.

The PS asks for six impaired scenarios; S0 is the added healthy baseline, not a seventh impairment. All seven conditions use 20 repetitions per browser.

4.3 Client and measurement environment

Component	Recorded campaign environment
Operating system	Ubuntu 24.04.5 LTS on x86_64 WSL2 host
Kernel	6.18.54.1-microsoft-standard-WSL2
Python	3.12.3
Browsers	Google Chrome 156.0.8078.73; Mozilla Firefox 159.0
Web server	Nginx 1.24.0
Capture / dissection	tcpdump 4.99.4 / TShark 4.2.2
Impairment controls	nftables 1.0.9; iproute2 / tc netem

The environment values above are transcribed from the campaign runbook/report record. They describe the host used for this campaign; they are not a compatibility guarantee for other machines.

4.4 Instrumentation and trial lifecycle

1. Preflight verifies topology, addresses, DNS, daemon health, route/impairment baseline and client reachability.

2. A unique trial hostname/profile is prepared; the selected scenario is applied and independently checked.
3. PCAP capture begins on the client link; the selected browser makes one HTTP/1.1 navigation and produces browser diagnostic telemetry.
4. Capture is stopped; the scenario is cleared; baseline restoration and profile cleanup are verified.
5. The runner writes a per-trial result and updates the campaign manifest atomically. The cycle repeats sequentially across the matrix.

4.5 Outcome classification and primary flow audit

The final interpretation identifies the primary GET / HTTP request and links its response to the TCP stream carrying that request. The raw PCAP audit inspected all 280 captures and found the main request in every capture. The address family derived from that primary flow matched the browser-reported winning family in all 280 cases. This protects the S5 analysis from false fallback classification caused by later favicon traffic or speculative preconnect.

Raw packet audit

280/280 primary GET / flows found • 0 missing • 0 main-flow/browser winning-family mismatches. The original PCAPs remain unchanged; the audit is a derived cross-check.

4.6 Campaign accounting and evidence quality

Measure	Campaign result
Planned/completed matrix slots	280 / 280
Total attempts / valid live trials	282 / 280
Failed attempts preserved / simulations	2 / 0
HTTP 200 in valid trials	280 / 280
PCAPs / packets	280 / 36,225
Captures reporting packet drops	0 / 280
PCAP checksum matches	280 / 280
Complete evidence index	1,410 archive files + 1 separately delivered campaign log = 1,411 indexed artifacts

Two initial cleanup failures remain in the audit trail and were retried under new unique trial IDs: Firefox S0 failed a preflight baseline self-test with a broken pipe; Chrome S6 failed cleanup because a helper operation required root privileges. They are excluded from the 280 valid measurements.

4.7 Results: winning address family

Scenario	Chrome IPv6	Chrome IPv4	Firefox IPv6	Firefox IPv4
S0	20	0	20	0
S1	0	20	0	20
S2	0	20	0	20
S3	20	0	20	0
S4	0	20	1	19
S5	18	2	15	5
S6	20	0	20	0
Total	98	42	56	84

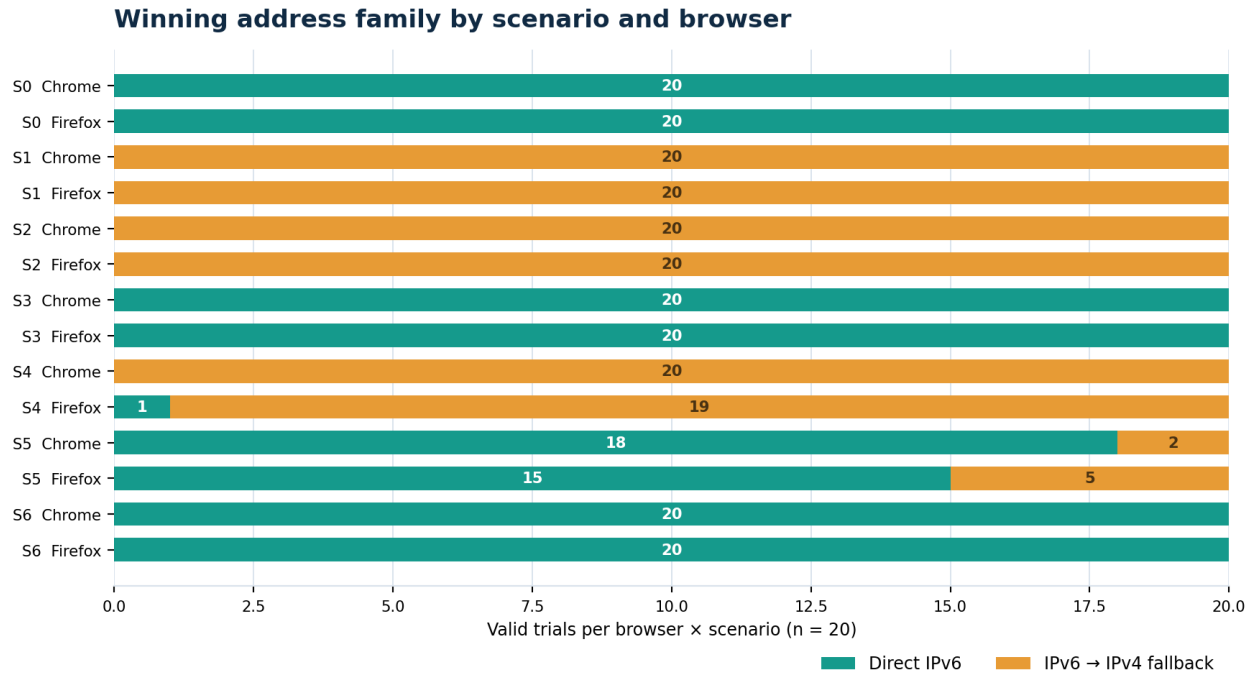


Figure 2. Direct IPv6 versus IPv4 fallback counts by scenario and browser.

Overall, 154/280 primary navigations (55.00%) completed over IPv6 and 126/280 (45.00%) over IPv4 after an IPv6 attempt. Chrome recorded 98 direct IPv6 and 42 IPv4-winning navigations; Firefox recorded 56 direct IPv6 and 84 IPv4-winning navigations. The different scenario outcomes should not be reduced to a browser-wide preference statistic: each browser was exposed to the same mix of controlled conditions.

4.8 Results: primary-document TCP handshake-response interval

Table values below are calculated from the raw PCAP flow carrying the primary GET /. The reported interval is client SYN to matching server SYN/ACK on that flow. IQR is Q3-Q1. The bootstrap CI is a deterministic percentile interval for the median (B=1,000, seed=42).

Chrome: main-flow packet timing (ms)

Scenario	N	Median	IQR	p95	95% median CI
S0	20	20.32	0.07	20.81	20.28-20.34
S1	20	20.96	0.45	21.63	20.75-21.14
S2	20	20.31	0.10	20.79	20.27-20.35
S3	20	150.65	0.35	154.48	150.58-150.81
S4	20	20.74	0.72	22.88	20.54-21.16
S5	20	20.46	0.25	20.71	20.36-20.56
S6	20	20.64	0.50	21.33	20.49-20.93

Firefox: main-flow packet timing (ms)

Scenario	N	Median	IQR	p95	95% median CI
S0	20	20.42	0.16	20.81	20.36-20.48
S1	20	20.41	0.87	29.03	20.37-20.92
S2	20	20.38	0.24	21.54	20.32-20.53
S3	20	150.39	0.21	150.91	150.32-150.47
S4	20	20.36	0.15	52.24	20.32-20.46
S5	20	20.50	0.22	22.43	20.40-20.59
S6	20	20.42	0.30	21.33	20.32-20.61

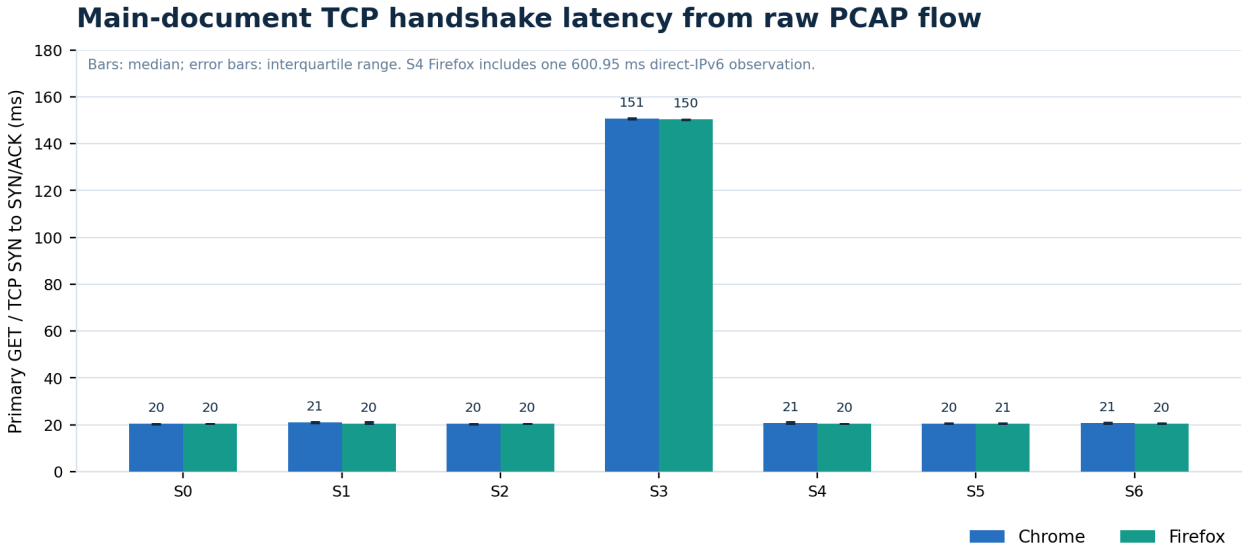


Figure 3. Median main-document SYN-to-SYN/ACK interval with IQR bars.

The configured RTTs are clearly expressed in the packet-derived measurements: S3 centres around 150 ms in both clients. S0/S1/S2/S4/S5/S6 main navigation flows generally return SYN/ACK at about 20 ms when IPv4 or a normal-RTT IPv6 path wins. In S4 Firefox, one direct IPv6 observation reaches 600.95 ms while 19/20 trials fall back to IPv4; the median therefore remains near 20 ms while the maximum is 600.95 ms.

4.9 Results: first inter-family SYN timing

Scenario / browser	N with interval	Median (ms)	IQR (ms)	Interpretation
S1 Chrome	20	300.83	0.53	IPv6 SYN blackhole; IPv4 attempt follows IPv6 timeout pacing.
S1 Firefox	20	408.45	30.43	IPv6 SYN blackhole; longer observed alternate-family pacing.
S2 Chrome	20	0.71	0.46	All 20 observed intervals are 0.35-6.82 ms; all below 10 ms.
S2 Firefox	20	160.43	45.66	All 20 observations are 130.78-336.23 ms.
S4 Chrome	20	300.81	0.72	IPv6 RTT 600 ms; IPv4 becomes the main-navigation winner.
S4 Firefox	19	397.94	40.12	19 fallbacks; one IPv6 direct trial did not have an IPv4 attempt interval.
S5 Chrome	5	300.96	0.90	Only five IPv4-first-SYN intervals; check each primary flow before calling fallback.
S5 Firefox	7	392.86	22.52	Only seven IPv4-first-SYN intervals; five other IPv4 sockets are ancillary/speculative in direct-v6 cases.

The first-SYN interval is descriptive of packet attempt pacing, but it is not itself a fallback success measure. In S5, some IPv4 traffic was triggered after the main IPv6 document navigation completed. The classification is therefore based on the primary HTTP flow, not simply on whether both address families appear anywhere in the capture.

4.10 S6 DNS Resolution Delay

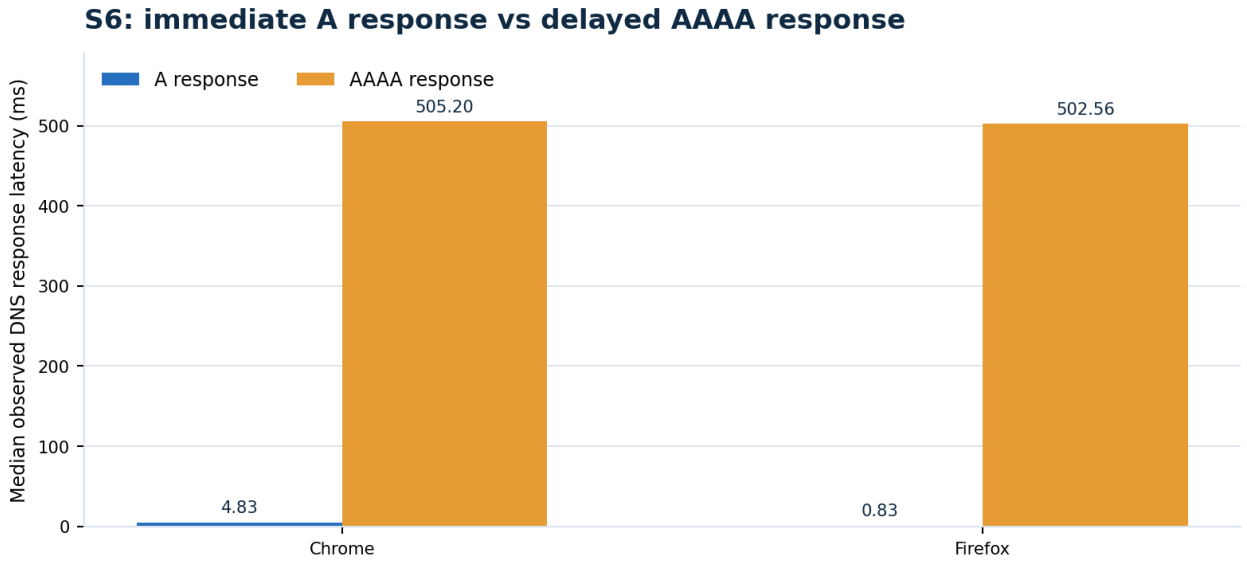


Figure 4. Recorded DNS A versus AAAA response latency in S6.

In S6, the DNS responder returns A immediately and delays AAAA by 500 ms. Median A response latency was 4.83 ms for Chrome and 0.83 ms for Firefox; median AAAA latency was 505.20 ms and 502.56 ms respectively. All 40 primary navigations completed over IPv6 after the delayed AAAA response was received. In these tested builds, no IPv4 primary navigation was observed after the recommended 50 ms Resolution Delay interval.

RFC 8305 section correction

Resolution Delay belongs to RFC 8305 Section 3 (Hostname Resolution Query Handling), not Section 4. Section 4 is address sorting. This report uses Section 3 throughout.

5. Optimization Proposed

The most material improvement during final analysis was not a new browser algorithm. It was an evidence-classification improvement: distinguish the primary navigation flow from subsequent ancillary connections, and make the packet-level result reproducible from raw captures.

Area	Earlier approach / risk	Final approach / improvement
Winning-family attribution	Treating any successfully established connection on both families as fallback could mislabel post-navigation traffic.	Correlate the primary GET / request and its HTTP response with one TCP stream; use that flow family to interpret the main navigation.
PCAP validation	Browser telemetry or first observed HTTP response can be insufficient when more than one HTTP request exists.	Re-parse all 280 PCAPs and compare the primary-flow family with the recorded browser winner: 280 found, zero mismatches.
Timing definition	Generic stored connection setup field can differ from the main-document packet event interval.	Publish a separately defined SYN-to-matching-SYN/ACK metric for the primary GET / flow; retain generic fields as separate metrics.
Reproducibility	Evidence and summaries can be hard to hand over as disconnected files.	Campaign manifest, results, PCAP indexes, two archives and a 1,411-entry evidence checksum index are preserved.
Offline analysis	A user may need Linux lab dependencies just to recompute summary tables.	Portable bundle contains the analyzer and valid trial JSON records; the Windows offline analyzer reproduced the delivered CSV exactly.

This is an optimization of measurement attribution and evidence engineering. The study did not optimize the internal code of Chrome or Firefox and makes no claim to have changed their Happy Eyeballs implementations.

6. Architecture Design (UML / Component View)

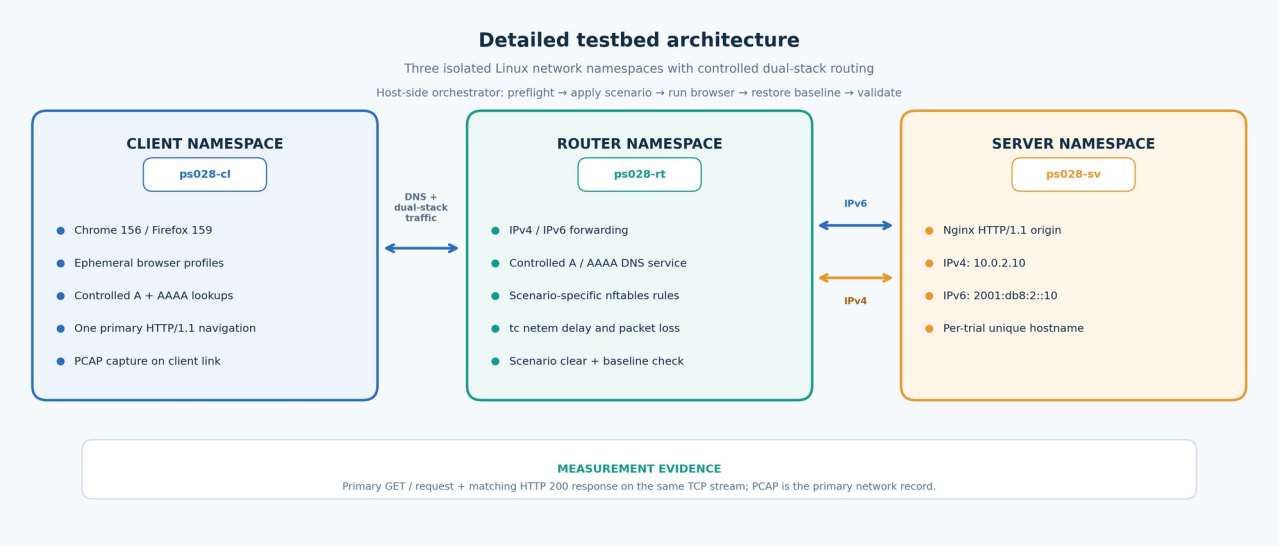


Figure 5. Detailed three-namespace architecture and dual-stack network paths.

The runtime components are the campaign orchestrator, scenario controller, browser driver, DNS service, router namespace, Nginx origin, tcpdump capture process and trial-result writer. The campaign runner serializes matrix execution and associates each slot with a unique trial identifier. The trial identifier links manifest entries, result files, PCAP files and browser diagnostic output.

7. Data Flow Diagram

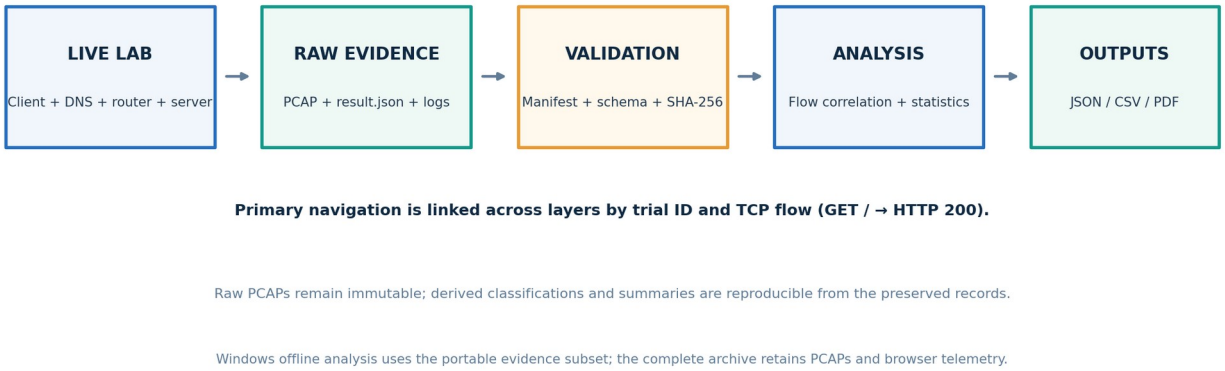


Figure 6. From live measurements to validated report artifacts.

The campaign manifest is the ledger of planned slots, attempts, completion state and exclusions. Per-trial result records store the observed winner, status and selected metrics. PCAPs preserve packet events, while browser NetLog/Firefox telemetry act as supporting diagnostics. The offline analysis engine ingests valid live records, keeps exclusions separate, computes statistics and writes JSON/CSV outputs. The additional raw-PCAP audit ties the main document request to its TCP stream and validates the family used by the primary navigation.

8. Deployment Timeline



Figure 7. Campaign execution and post-campaign verification sequence.

Campaign start: 29 Oct 2026 20:54:25 UTC; completion: 29 Oct 2026 22:57:12 UTC; elapsed time: 122.79 minutes (2 h 2 min 47.1 s).

The manifest records 282 attempts for 280 planned slots. Two first attempts encountered cleanup/preflight failures and were retried using new trial IDs; the campaign completed with no pending slots and no simulation entries. After collection, the evidence was indexed, checksum-verified, analysed offline, re-audited at the main HTTP-flow level and checked against the recorded 400-test execution log.

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10. Closing

10.1 Conclusion

The campaign met the core experimental dimensions of PS-028: two existing clients, six controlled impairment scenarios, a healthy baseline, 20 repetitions per condition/browser, controlled DNS, trial-level records, packet captures, winning-family analysis and connection timing comparisons. In the observed testbed, both browsers used IPv6 in the healthy baseline and high-but-not-extreme IPv6 RTT scenario; both transitioned to IPv4 in every IPv6 SYN-blackhole and TCP-RST case; outcomes were mixed under 50% IPv6 SYN loss and nearly all IPv4 under the 600 ms IPv6 RTT condition. Under delayed AAAA, both waited for the delayed IPv6 address and then used IPv6.

The additional raw-PCAP re-audit strengthens the central claim: the winning address family was tied to the primary HTTP document request, rather than inferred from any later connection. All 280 captures contained a matching primary flow and the audited address family agreed with the recorded browser result. These findings are specific to the tested browser builds, host and HTTP/TCP setup.

10.2 Limitations and threats to validity

- Only one recorded x86_64 Ubuntu 24.04.5 / WSL2 laboratory host and two browser builds were measured. Results should not be generalized to all browsers, releases, operating systems or networks.
- The campaign measured HTTP/1.1 over TCP. It did not evaluate HTTPS/TLS handshake timing, QUIC/HTTP/3, SVCB/HTTPS records or the full HEv3 feature set.
- Ephemeral profiles isolate trials but do not test persistent cross-session failure memory or adaptive statefulness.
- Browser telemetry is diagnostic and version-specific. Packet captures are the primary network evidence.
- The generic stored connection_setup_time_ms field differed by more than 1 ms from the main-flow SYN-to-SYN/ACK interval in 144/280 trials. The final report therefore defines and uses the explicit raw-PCAP primary-flow timing metric.
- The response_body_bytes metric is the length of browser process stdout, not HTTP payload size (308 bytes Chrome; 72 bytes Firefox). Browser-duration output mechanisms also differ and should not be treated as a like-for-like readiness comparison.

- The supplied rfc8305_assessment.csv retains stale/misleading S2/S6 assessment text. This report interprets those cases from the per-trial records and packet observations; regenerate the auxiliary CSV before using it as a standards assessment.
- The official Proposed-structure-hackathon.pdf template was not included among supplied files. This report follows the 11-section sequence documented in the study guide G12 mapping; a later organizer-provided template should be checked.

10.3 Reproducibility and submission checklist

Item	Status / location
Scenario runner and canonical config	Implemented: lab/run_campaign.py, lab/run_trial.py, lab/scenarios.yaml
Two browser clients	Recorded builds: Chrome 156.0.8078.73 and Firefox 159.0
Six impaired scenarios + baseline	S0-S6, 20 repetitions per browser/condition
Trial records and PCAP evidence	280 valid result files and 280 PCAPs in campaign directory
Checksum / evidence package	1,411 indexed artifacts: 1,410 in complete TAR plus separate campaign log
Offline analysis	Portable analyzer re-run; supplied CSV reproduced exactly
Test execution	Recorded local run: 400 tests in 20.927 s; final OK
PDF report upload	Prepared: docs/PS028_Final_Experimental_Report.pdf
Mentor prototype + pseudocode demonstration	Still a submission activity; introduce demo sequence and show pseudocode below
Repository administrative condition	PS asks for private team-named repository and aiori-hackathon collaborator/ownership process; confirm with organizers

10.4 Prototype pseudocode

```
for scenario in [S0, S1, S2, S3, S4, S5, S6]:
    for browser in [Chrome, Firefox]:
        repeat 20 times:
            assert baseline_is_clean()
            trial = create_unique_trial_id(scenario, browser)
            apply_scenario(scenario)
            assert impairment_self_test(scenario)
            start_client_link_pcap(trial)
            run_one_fresh_browser_navigation(browser, unique_hostname(trial))
            stop_pcap_and_collect_browser_telemetry(trial)
            restore_baseline()
            assert baseline_is_clean()
            save_result_and_update_manifest(trial)

for trial in valid_live_trials(manifest):
    main_flow = correlate_primary_GET_slash_and_HTTP_200_in_PCAP(trial.pcap)
    assert main_flow.address_family == trial.browser_winner_family
    derive_primary_SYN_to_SYNACK_interval(main_flow)

write_summary_json_csv_and_report()
```

Final statement

The evidence supports a reproducible, controlled comparison of these two browser builds. It does not establish universal behaviour or full RFC conformance. The final report should be read together with the manifest, per-trial results, PCAP archive, checksum indexes and the recorded local test log.